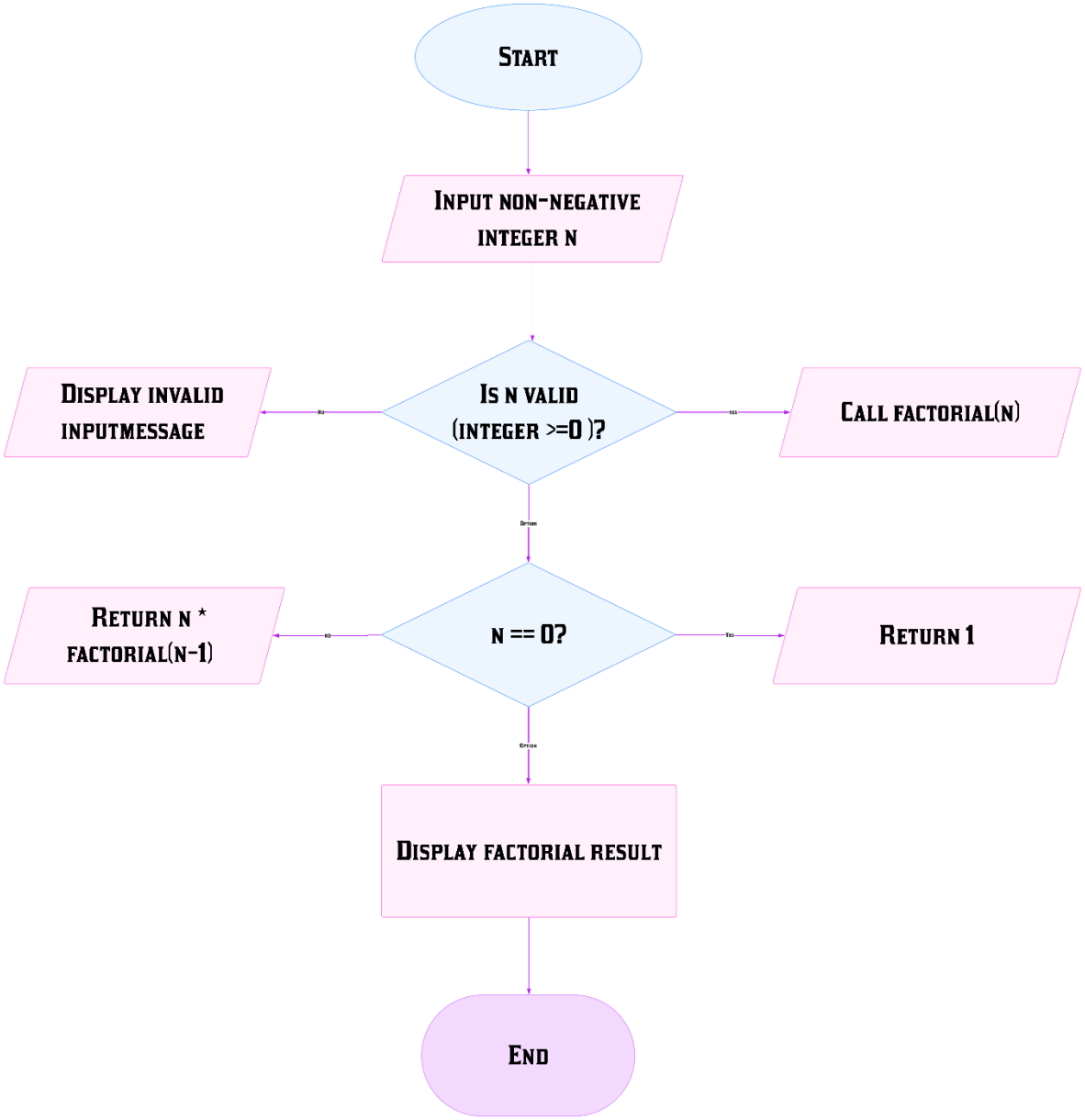
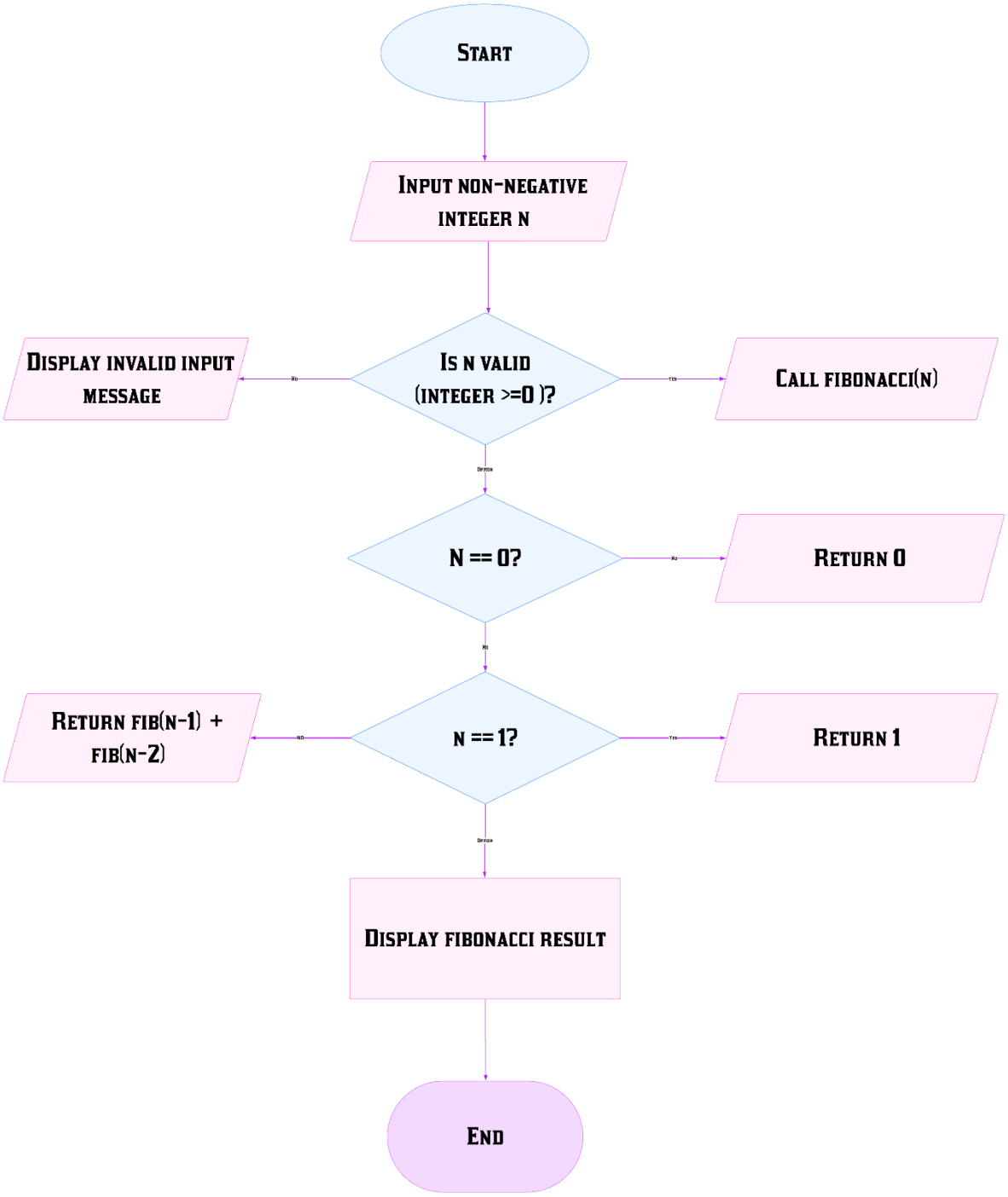
RECURSION ACTIVITY

FACTORIAL FLOWCHART



FIBONACCI FLOWCHART



FACTORIAL PSEUDOCODE

START

```
  FUNCTION factorial(n)
    IF n < 0 THEN
      RETURN "Invalid input"
    END IF
    IF n = = 0 THEN
```

```
        RETURN 1
    ELSE
        RETURN n * factorial(n – 1)
    END IF
  END FUNCTION
END
```

FIBONACCI PSEUDOCODE

```
START
   FUNCTION fibonacci(n)
      If n < 0  THEN
          RETURN "Invalid input"
    END IF
    IF n == 0 THEN
          RETURN 0
    ELSE IF n == 1 THEN
           RETURN 1
    ELSE
           RETURN fibonnaci(n -1) + fibonnaci(n – 2)
    END IF
   END FUNCTION
END
```

PYTHON PROGRAM

```
# Recursion Program: Factorial and Fibonacci
# Factorial Function
def factorial(n):
   if n < 0:
      return "Invalid input! Number must be non-negative."
   elif n == 0:
```

```
        return 1
    else:
        return n * factorial(n – 1)
# Fibonacci Function
def fibonacci(n):
    if n < 0:
        return “Invalid input! Number must be non-negative.”
    elif n == 0:
        return 0
    elif n == 1:
        return 1
    else:
        return fibonacci(n -1) + fibonacci(n-2)
# Main Program
try:
    num = int(input(“Enter a non-negative integer:”))
    print(“\nFactorial Result:”)
    print(f”{num}! =”,factorial(num))
    print(“\nFibonacci Result:”)
    print(f”Fibonacci({num}) =”fibonacci(num))
except ValueError:
    print(“Invalid input! Please enter an integer.”)
```

THE END